A Multimodal-based Framework for Smart ECG Report Generation

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Abstract. An electrocardiogram (ECG) is a vital instrument for heart-related disease diagnosis through wearable devices for early treatment. Despite the recent rapid advancement in the development of large language models (LLMs), they are not widely adopted and utilized. In this work, we propose a multimodal framework for ECG signal processing and interpretation based on LLMs and vision-based language models (VLMs). By aligning the ECG time series features, text-based clinical ECG report, and visualized ECG signal through a trainable attention module, our framework provides accurate physiological analysis. Our findings demonstrate that by leveraging pre-trained LLMs and with the addition of the visual domain, which is rarely used in ECG diagnosis, we can enhance medical text interpretation without a heavy retraining computational load. This work contributes toward a more efficient and accurate physiological diagnosis, which enables smart AI healthcare applications and a potential multi-task medical assistant in the future.

Keywords: Electrocardiogram · Multimodal AI · Large Language Models · AI in Healthcare · Report Generation.

1 Introduction

Electrocardiograms (ECGs) play a crucial role in assessing a patient's physiological states by capturing the cardio signal and monitoring the heart's function. Nevertheless, ECG signals are generally susceptible to noise and interferences such as electrode misalignment, movement artifacts, or electrical noise during recording sessions or through wearable sensors, which can affect the morphological characteristics of the waveforms, causing misreading or misdiagnosis based on the received signals [4]. In addition, it is time-consuming, inefficient, and cost-ineffective to manually extract and process the hand-crafted ECG features. Thus, automated ECG signal processing becomes the preferred choice due to superior accuracy and lower inference time. Traditional ECG processing methods rely on conventional threshold-based and rule-based approaches, which cannot adapt to complex, non-linear ECG artifact patterns; thus, their capabilities to deal with practical signals are limited [7]. Recently, deep learning-based ECG signal processing has gained momentum due to the rapid advancement of convolutional neural networks (CNNs) and transformer-based processing. Using deep layers of large parameter convolutional blocks, they improve the feature extraction process, extract more meaningful information from the input ECG waveforms, and increase the classification accuracy [3, 5]. In addition, with the breakthrough in natural language processing (NLP), particularly large language models (LLMs), several studies explore the alignment between textual features and extracted ECG features to generate diagnosis and interpretation [1]. However, these works mostly focus on feature alignment between the textual and single-dimensional (1-D) ECG domains and neglect other modalities.

In this work, we propose a multimodal framework that leverages the visual, textual, and physiological domains to accurately generate textual reports for ECG signal analysis. Experiment results prove that by utilizing pre-trained large vision language models (LVLMs) as the visual encoder, and pre-trained LLMs as the output head for textual generation, we can achieve modestly accurate ECG interpretation, enabling future research on multimodal ECG signal processing for smart artificial intelligence (AI) applications in biomedical and healthcare.

2 Methodology

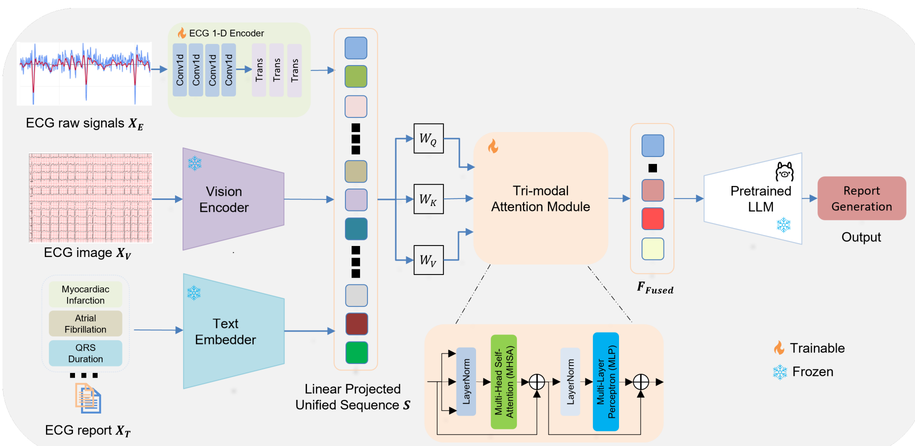

Fig. 1. Architecture of our proposed framework with the input raw ECG signals X_E, visualized ECG X_V, and the textual report X_T. The unified sequence S is produced by concatenating the linear projected embeddings from the extracted features of the three encoders, and F_{Fused} acts as the input fed into the pre-trained LLM to produce natural language generation output.

Our proposed tri-modal framework is summarized in Figure 1. The architecture consists of a lightweight hybrid CNN-Transformer ECG encoder, text embeddings, a pre-trained vision encoder, a multimodal attention mechanism that utilizes the unified sequence to model the intra-domain and cross-domain relationships, and a pre-trained LLM that serves as the task head for text generation. Let the input for the 3 encoders be X_E, X_V, and X_T, respectively.

ECG Encoder. To process the 12-lead ECG signals, a hybrid architecture of CNN and Transformers was used to capture both local morphological features and long-range dependencies of the signal. The signal first goes through four layers of 1-D CNN to capture the local waveform patterns, with the last layer expanding to the channel dimension D_E. Subsequently, three Transformers blocks, each consisting of a Multi-Head Self-Attention (MHSA) module with a Feed-Forward Network (FFN), were used to produce the ECG feature F_E throughout the recording duration N_E defined as

$$F_E = E_{ECG}(X_E) = \text{Transformer}(\text{CNN}(X_E) + P_{pos}), \quad F_E \in \mathbb{R}^{N_E \times D_E}. \tag{1}$$

Text Embedder. To extract the information from each ECG report, we employ the pre-trained text embedder [6] to map tokenized clinical texts into a sequence of dimensional embeddings, D_T. By leveraging a pre-trained text encoder from a vision-language model, we preserve the fine-grained details that are already aligned with visual input, accelerating the training of the subsequent fusion module, shown as: $F_T = E_T(X_T), \quad F_T \in \mathbb{R}^{N_T \times D_T}$.

Vision Encoder. Capturing high-level features from the visualized ECG image is performed via a pre-trained vision encoder. While ECG images differ significantly from natural images, utilizing a pre-trained encoder on natural images (e.g., ImageNet) is highly beneficial. The shallow convolutional layers act as generalized feature extractors, detecting low-level features such as grid lines, waveform morphologies (e.g., P-waves, QRS complexes), and signal spikes. Given the input image X_V, the pre-trained encoder E_V produces the output visual features F_V as: $F_V = E_V(X_V), \quad F_V \in \mathbb{R}^{N_V \times D_V}$.

Multimodal Attention. Unlike pairwise cross-attention (text-image, text-ECG signal) that scales quadratically with the number of modalities, we first employ a linear projection on each domain output feature F_i ($i \in \{E, T, V\}$) from each encoder to a shared embedding space D_{shared} to obtain the unified feature $\hat{F}_i = F_i W_i + b_i$, where $W_i \in \mathbb{R}^{D_i \times D_{shared}}$. By utilizing modality embedding U_i, the three feature embeddings $H_i = \hat{F}_i + U_i$, where $U_i \in \mathbb{R}^{1 \times D_{shared}}$, are then concatenated together to form a single unified sequence $S = \text{Concat}(H_V, H_T, H_E)$, where $S \in \mathbb{R}^{(N_V+N_T+N_E) \times D_{shared}}$. A Transformer Self-Attention layer is then used to model the intra-domain relationship, e.g., 1-D ECG and 1-D ECG, and cross-domain relationship accordingly to produce the attention Z_{Fused}:

$$Z_{Fused} = \text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{D_{shared}}}\right), \tag{2}$$

where $Q = SW_Q$, $K = SW_K$, and $V = SW_V$ are the query, key, and value respectively. Finally, a LayerNorm layer is used to produce the fused feature F_{Fused} to the desired output dimension for the LLM input: $F_{Fused} = \text{LayerNorm}(S + Z_{fused})$.

Pre-trained LLM as Task Head. To translate the output multimodal fused features into an ECG textual report, we utilize state-of-the-art pre-trained LLMs and prompt them to generate a natural language description of the ECG signal. All prompts are embedded to form a prompt representation, concatenated with the final output multimodal ECG features to form the input for the LLMs, described by $O_{LLM} = \text{MLP}(F_{Fused}), \quad O_{LLM} \in \mathbb{R}^{(N_V+N_T+N_E) \times D_{LLM}}$.

Table 1. Evaluation of natural language generation metrics on the MEETI dataset. Model sizes are denoted by 'M' for millions and 'B' for billions of parameters. Results highlighted in **Bold** and *Underline italic* represent the best and second-best scores, respectively.

Vision Encoder	Size	Interpreted LLM	Size	BLEU-1	BLEU-2	BLEU-4	ROUGE-L	ROUGE-1	ROUGE-2
LLaVA-Next	7B	LLama-3	8B	*0.722*	*0.687*	*0.625*	*0.759*	*0.780*	*0.683*
LLaVA-Next	7B	Mistral-Nemo	12B	**0.745**	**0.701**	**0.658**	**0.773**	**0.786**	**0.696**
SigCLIP	400M	LLama-3	8B	0.632	0.538	0.461	0.556	0.601	0.574
SigCLIP	400M	Mistral-Nemo	12B	0.648	0.572	0.491	0.583	0.628	0.602
DINOv2-Giant	1.1B	LLama-3	8B	0.651	0.582	0.502	0.637	0.673	0.601
DINOv2-Giant	1.1B	Mistral-Nemo	12B	0.682	0.639	0.579	0.710	0.701	0.673
BiomedCLIP	200M	LLama-3	8B	0.629	0.535	0.479	0.548	0.593	0.565
BiomedCLIP	200M	Mistral-Nemo	12B	0.636	0.535	0.449	0.573	0.612	0.593

3 Experiments

3.1 Experiment Settings

In this work, the MIMIC-IV dataset [2] and its extension MEETI [8] are used for the visualized ECG images. MIMIC-IV is the largest publicly available multi-modal ECG dataset with 12-lead ECG raw signals with corresponding diagnostic reports, while the MEETI dataset provides raw ECG images corresponding to MIMIC-IV. We use 85% of the dataset for training and evaluate the textual generation ability on the last 15% of the dataset. The experiments are conducted on a single NVIDIA RTX 5090 GPU with PyTorch 2.0. AdamW optimizer is used with a learning rate (LR) of 10^{-3}, $\beta_1 = 0.9$, and $\beta_2 = 0.99$. For the learning rate scheduler, CosineAnnealingLR is used with a minimum LR of 10^{-5}. Batch size is set to 32 with 100 training epochs, early stopping is employed after 5 epochs without improvement in validation loss, and the dropout rate is set to 0.1 with 10^{-4} weight decay to prevent overfitting.

3.2 Results and Discussion

To assess the quality of the generated ECG report, we use the two common natural language generation metrics, BLEU and ROUGE. BLEU (Bilingual Evaluation Understudy) is used to measure phrase overlap (n-grams) between the generated report and the clinician's ground-truth report. On the other hand, ROUGE (Recall-Oriented Understudy for Gisting Evaluation) focuses on the recall rate of how much the clinical texts is captured by the model, with ROUGE-L measures the longest common subsequence. Table 1 illustrates the experiment results using 4 different vision encoders: LLaVA-Next, SigCLIP, DINOv2, and BiomedCLIP; and 2 different LLMs for output interpretation, LLama-3 and Mistral-Nemo.

From Table 1, it can be seen that combining LLaVA-Next and Mistral-Nemo yields the best results across all metrics. This can be potentially attributed to the complexity of the models (7B parameters and 12B parameters, respectively), proving the superior ability to capture and encode high-level information and complex structural dependencies compared to smaller-scale encoders. In addition, BiomedCLIP (200M parameters), a biomedical visual encoder, yields almost similar performance (e.g., BLEU-1 of 0.636 vs. 0.648 with Mistral Nemo) to SigCLIP (400M parameters), an encoder with twice the number of parameters, proving the usefulness of domain-specific pre-training. Nevertheless, these encoders are more efficient in terms of computational complexity compared to the state-of-the-art methods and yield viable results, which make them more suitable for a resource-constrained environment or mass deployment.

4 Conclusions

In this work, we propose a novel tri-modal ECG report generation framework integrating the raw 1-D signals, visualized images, and clinical textual reports. Extensive experiment results across multiple encoder and decoder configurations demonstrate the framework's ability to generate accurate diagnosis reports. In the future, we plan to expand our framework to a multi-task system that is capable of ECG signal quality assessment and denoising, and utilize non-linear extracted ECG features to further improve the framework's performance, contributing to the advancement of AI-based applications in the biomedical field.

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